

Retail Sales Performance Analysis

Data Profiling Report

Version: Final

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1. Purpose

This report summarizes the basic profiling, validation, and cleaning performed before the main business analysis. The focus was on making the data reliable enough for SQL analysis and Power BI reporting.

2. Dataset Summary

Field	Details
Project Name	Retail Sales Performance Analysis
Document	Data Profiling & Cleaning Report
Industry	Retail / E-commerce
Tools	Excel, PostgreSQL
Source	Kaggle
File Format	CSV
Number of Files	6

The dataset contains retail transaction data along with customer, product, and campaign information used for the analysis.

Main tables:

- fact_transaction
- dim_customer
- dim_product
- dim_campaign
- bridge_campaign_customer
- fact_coupon_redemption

3. Data Quality Assessment

The following basic checks were performed:

- Dataset size and structure
- Unique customers and products
- Transaction range
- Missing values
- Duplicate records
- Negative quantities
- Invalid sales values
- Blank or unknown categories

These checks helped identify issues that could affect the business analysis.

4. Data Quality Checks & Actions

Risk	Business Impact
Missing & Unknown Values	Blank and unknown values were reviewed, but valid unknown categories were retained where they did not affect the analysis.
Transaction Validation	Transaction records were reviewed for cancelled or unusual transactions before analysis.
Quantity & Sales Validation	Quantity and sales-related fields were checked for negative or unusual values.
Duplicate Records	Duplicate records were checked and confirmed duplicates were removed where necessary.

5. Data Cleaning Actions

The main cleaning actions included:

- Removing confirmed duplicates
- Reviewing invalid or unusual values
- Checking key customer and product fields
- Reviewing blank and unknown categories
- Validating campaign and redemption data

The original business structure of the dataset was preserved.

6. Validation Summary

After cleaning, the main analytical tables and key fields were rechecked.

The data was considered suitable for:

- SQL analysis
- Data modeling
- DAX calculations
- Power BI dashboard development

7. Data Readiness

The cleaned dataset was ready for the next stages of the project:

SQL Analysis → Data Modeling → DAX → Power BI Dashboard

8. Data Limitations

- Some fields contain blank or unknown values.
- Certain records require business context interpretation.
- The dataset is used for portfolio analysis rather than live production reporting.
- Advanced statistical data-quality techniques were outside the project scope.

9. Data Limitations

The data was profiled, validated, and cleaned using a basic business-focused approach.

After these checks, the dataset was considered suitable for the project's analysis and dashboard devel.

10. Conclusion

The dataset was profiled, validated, and cleaned using a basic business-focused approach. Key data quality issues were reviewed, and the cleaned data was considered suitable for SQL analysis, data modeling, and Power BI dashboard development.